

IP Addressing & Subnetting Notes

IP Addressing

What is an IP Address?

An IP address is a unique identifier for a device on a network.

IP = Device identity + Location on network

- **Example:** 192.168.1.10
- **Structure:** IPv4 = 32 bits, divided into 4 octets. Each part = 8 bits (values range from 0-255).
- **Binary Concept:** SOC analysts do not use binary daily, but understanding it helps in subnetting.

Network Part vs. Host Part

An IP address is split into two main sections:

1. **Network Part:** This identifies which network the device belongs to. Think of it like a **City or Area** in a physical address.
Example: In 192.168.1.10, the network part might be 192.168.1. This means all devices with 192.168.1.X are in the same network.
2. **Host Part:** This identifies the specific device inside that network. Think of it like a **House number** in a city.
Example: In 192.168.1.10, the host part is 10, representing device number 10 in that network.

Subnetting Basics

What is Subnetting?

Subnetting means dividing a network into smaller networks.

- **Why do we do it?** Better security, Better performance, and Easier management.

CIDR Notation

- **CIDR stands for:** Classless Inter-Domain Routing. It improves IP address management by allowing more flexible subnetting and efficient allocation, significantly reducing address waste.
- **Example:** 192.168.1.0/24
/24 means that 24 bits are dedicated to the network portion, leaving 8 bits available for the host portion.

How Many IP Addresses in a /24 Subnet?

A /24 subnet provides a total of 256 IP addresses.

- **Formula:** Total IPs = $2^{(32 - \text{subnet prefix})}$
- **Calculation:** Total IPs = $2^{(32 - 24)} = 2^8 = 256$

- **Usable IPs:** Out of these 256 IPs, **two are reserved**:
 1. **Network Address:** The first IP (e.g., 192.168.0.0), which identifies the subnet itself.
 2. **Broadcast Address:** The last IP (e.g., 192.168.0.255), used to communicate with all hosts in the subnet.
- Thus, a /24 subnet has **254 usable IP addresses** for assignment to devices.

Common Subnets Reference Table

CIDR	Subnet Mask	Usable Hosts
/24	255.255.255.0	254
/16	255.255.0.0	65,534
/8	255.0.0.0	16M

Versions of IP Addresses

- **IPv4 (Internet Protocol Version 4):**
 - Address size: 32-bit
 - Format: Decimal (e.g., 192.168.1.1)
 - Total addresses: 4,294,967,296
 - IPsec: Optional
 - Status: Widely used but facing address exhaustion
- **IPv6 (Internet Protocol Version 6):**
 - Address size: 128-bit
 - Format: Hexadecimal (e.g., 2001:db8::1)
 - Total addresses: 3.4×10^{38}
 - IPsec: Mandatory
 - Status: Designed to solve the IPv4 address shortage

Classes of IP Addresses

- **Class A:**
 - Range: 1.0.0.0 – 126.255.255.255
 - Bits: Network bits: 8, Host bits: 24
 - Use: Large networks (e.g., Government organizations)
 - *Note: 127 is reserved for loopback testing.**
- **Class B:**
 - Range: 128.0.0.0 – 191.255.255.255
 - Bits: Network bits: 16, Host bits: 16
 - Use: Medium-sized networks (e.g., Universities)
- **Class C:**
 - Range: 192.0.0.0 – 223.255.255.255

- Bits: Network bits: 24, Host bits: 8
- Use: Small networks (e.g., Home and small businesses)
- **Class D:**
 - Range: 224.0.0.0 – 239.255.255.255
 - Use: Multicast communication (e.g., Video streaming)
- **Class E:**
 - Range: 240.0.0.0 – 255.255.255.255
 - Use: Experimental and research (Not used for public networking)

Four Main Types of IP Addresses:

1. Public | 2. Private | 3. Static | 4. Dynamic

SOC Perspective on IP Types:

- **Private IP:** Internal traffic.
- **Public IP:** External connections.
- **SOC Example:** If a log shows Source: 192.168.1.10 and Destination: 45.77.123.66, it indicates **Internal** → **External communication**.

MAC Address (Media Access Control Address)

What is a MAC Address?

A MAC address is a physical, hardware-level unique identifier assigned to a Network Interface Controller (NIC) of a computing device.

- **Example:** 00:1A:2B:3C:4D:5E
- **Structure:** 48-bit address long (6 bytes), written in hexadecimal format.
- **Parts:**
 - First half → Organizationally Unique Identifier (OUI / Manufacturer ID).
 - Second half → Network Interface Controller Specific (Device ID).

ARP (Address Resolution Protocol)

- **What it does:** ARP maps an **IP address** → **MAC address**.
- **Example query:** "Who has 192.168.1.1? Tell me your MAC address."

MAC Address vs. IP Address Reference

Feature	MAC Address	IP Address
Full Form	Media Access Control Address	Internet Protocol Address
Alternative Names	Physical, hardware, or ethernet address	Logical address, network, or internet address

Feature	MAC Address	IP Address
Definition	Unique identifier assigned to the device's NIC	Numerical label assigned to devices connected to an IP network
Core Function	Helps to uniquely identify the physical device	Helps to identify the network connection of a device on the internet
Assignment	Assigned by the device manufacturer	Assigned by an IT administrator or an ISP
Flexibility	Cannot be changed	Can be changed
Length	48 bits (6 bytes) long	IPv4 - 32 bits (4 bytes) / IPv6 - 128 bits (16 bytes)
OSI Layer	Works in the Data Link Layer	Works in the Network Layer

Real SOC Scenario

Observation: Multiple devices showing the exact same MAC address on the network.

Analysis: This is highly likely an **ARP Spoofing / Man-In-The-Middle (MITM) attack**.